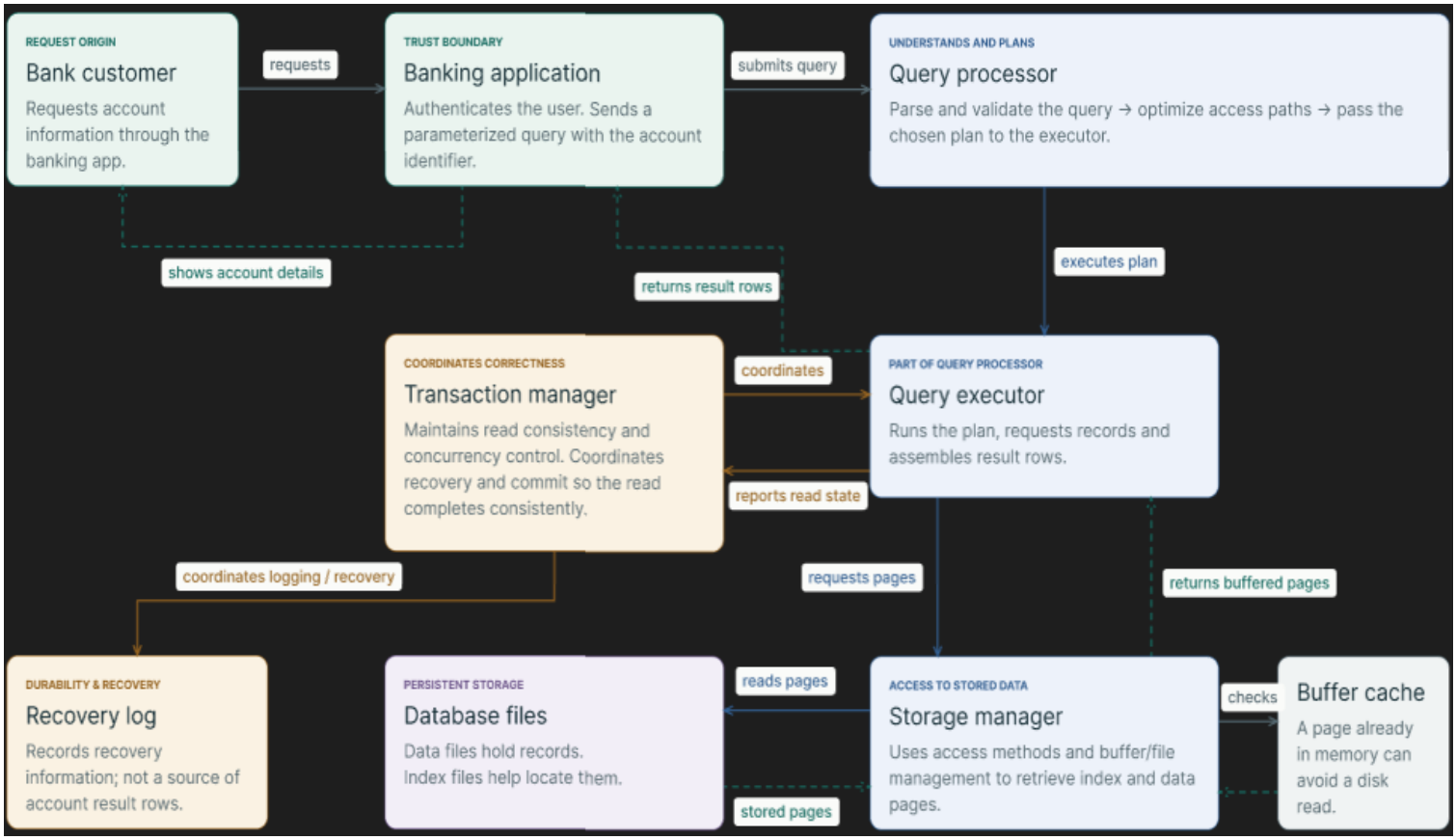
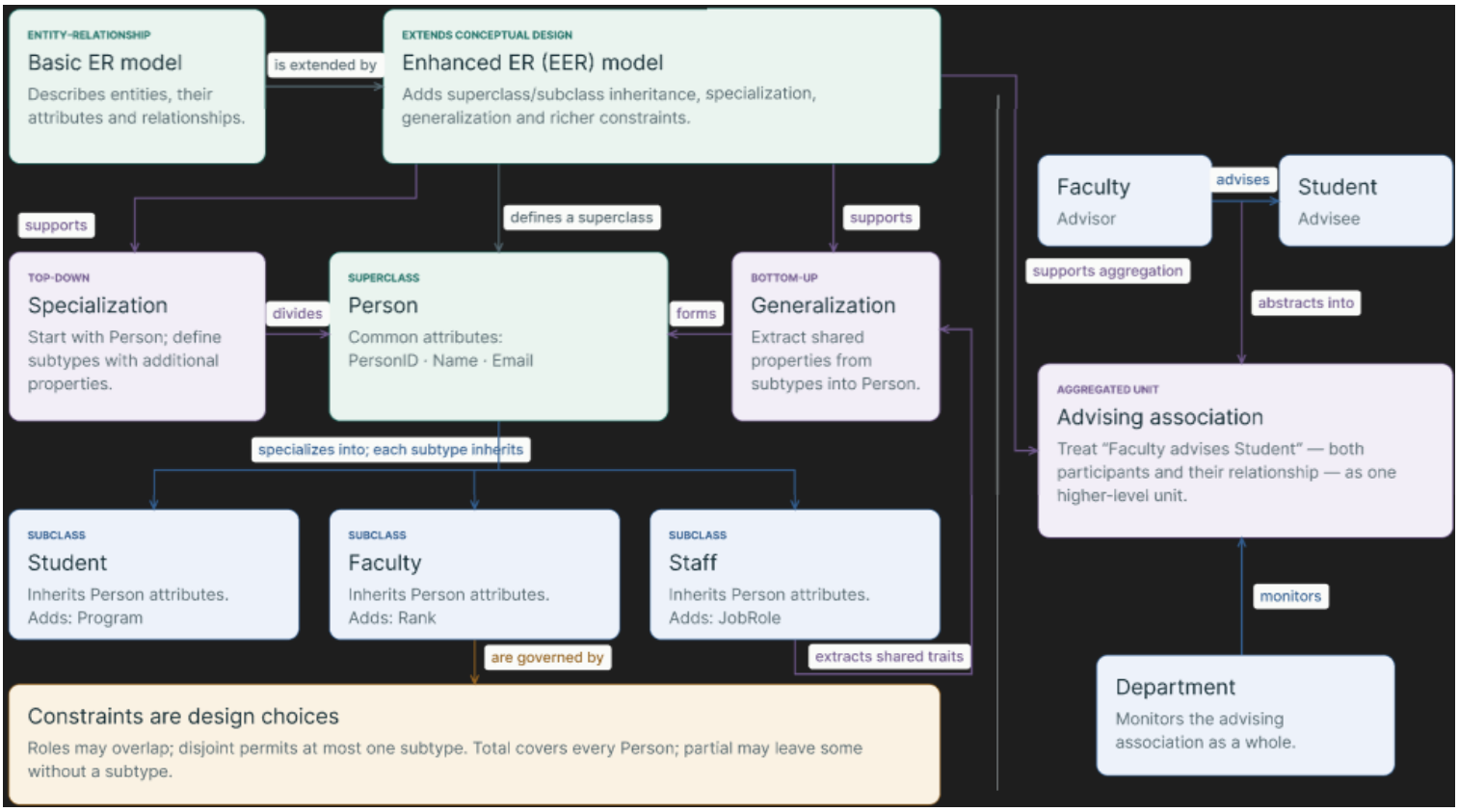
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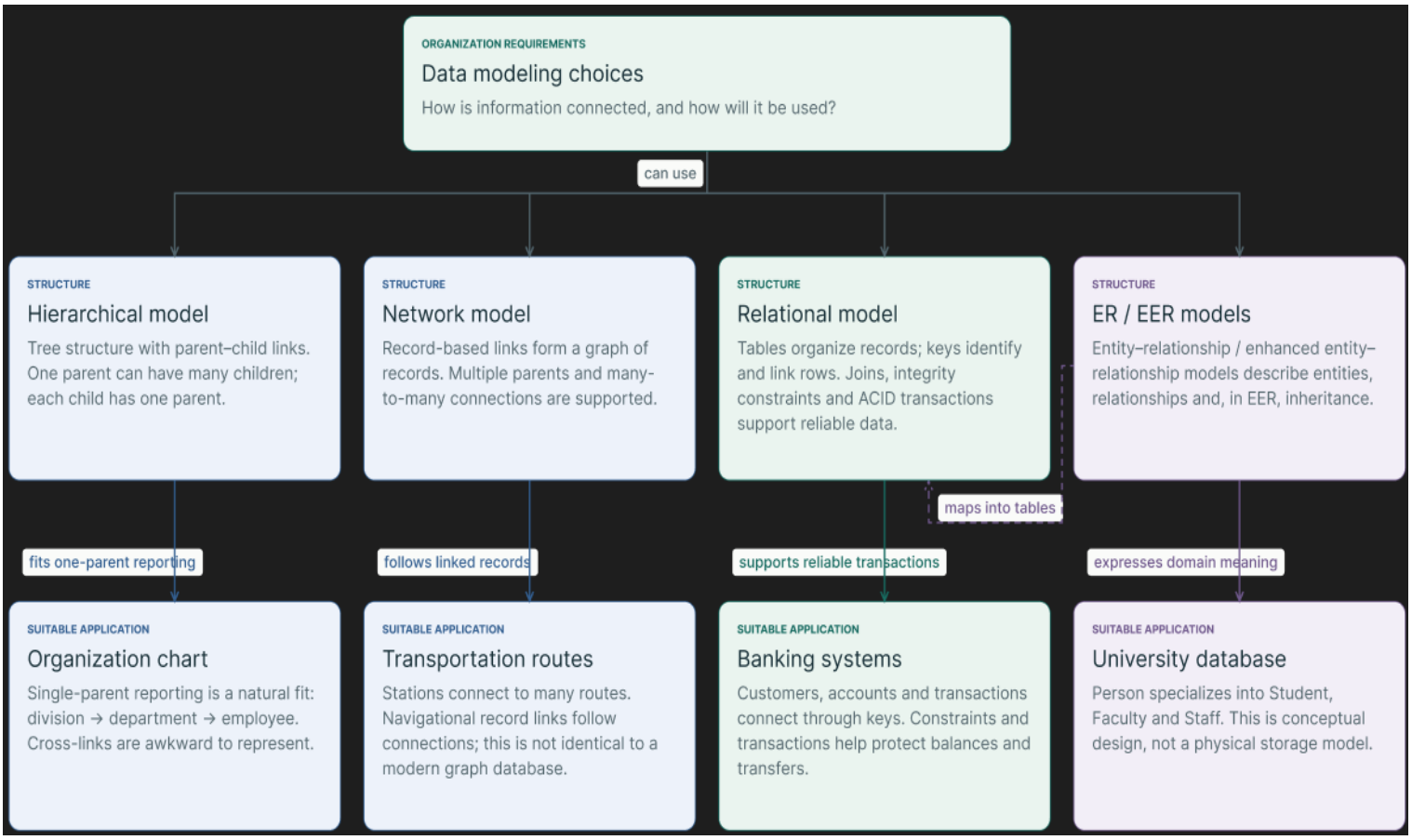
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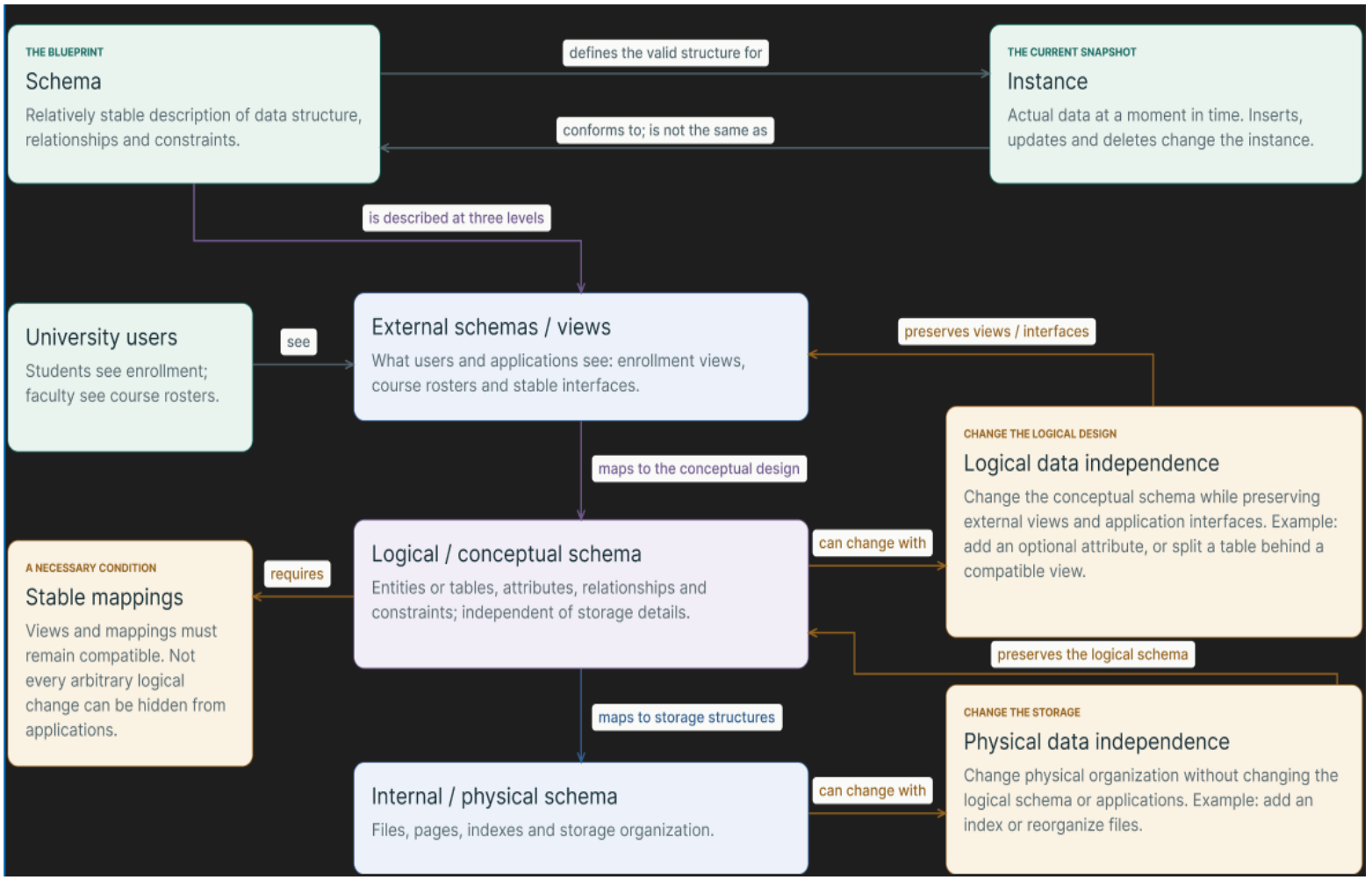
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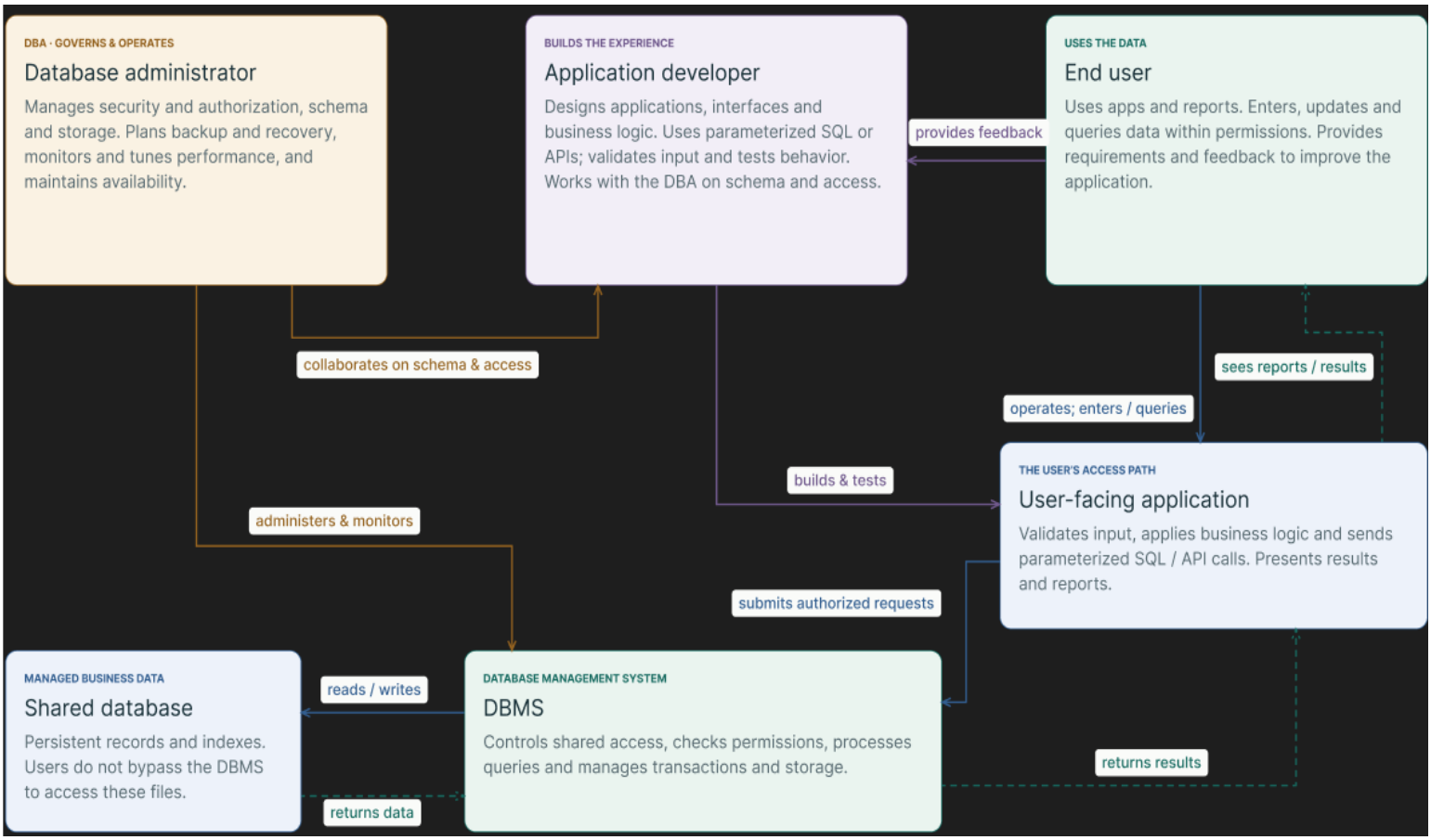
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ORGANIZATION REQUIREMENTS
Data modeling choices
How is information connected, and how will it be used?
can use
STRUCTURE
Hierarchical model
Tree structure with parent–child links. One parent can have many children; each child has one parent.
STRUCTURE
Network model
Record-based links form a graph of records. Multiple parents and many-to-many connections are supported.
STRUCTURE
Relational model
Tables organize records; keys identify and link rows. Joins, integrity constraints and ACID transactions support reliable data.
STRUCTURE
ER / EER models
Entity–relationship / enhanced entity–relationship models describe entities, relationships and, in EER, inheritance.
maps into tables
fits one-parent reporting
follows linked records
supports reliable transactions
expresses domain meaning
SUITABLE APPLICATION
Organization chart
Single-parent reporting is a natural fit: division → department → employee. Cross-links are awkward to represent.
SUITABLE APPLICATION
Transportation routes
Stations connect to many routes. Navigational record links follow connections; this is not identical to a modern graph database.
SUITABLE APPLICATION
Banking systems
Customers, accounts and transactions connect through keys. Constraints and transactions help protect balances and transfers.
SUITABLE APPLICATION
University database
Person specializes into Student, Faculty and Staff. This is conceptual design, not a physical storage model.


THE BLUEPRINT
Schema
Relatively stable description of data structure, relationships and constraints.
defines the valid structure for
conforms to; is not the same as
THE CURRENT SNAPSHOT
Instance
Actual data at a moment in time. Inserts, updates and deletes change the instance.
is described at three levels
University users
Students see enrollment; faculty see course rosters.
see
External schemas / views
What users and applications see: enrollment views, course rosters and stable interfaces.
preserves views / interfaces
maps to the conceptual design
CHANGE THE LOGICAL DESIGN
Logical data independence
Change the conceptual schema while preserving external views and application interfaces. Example: add an optional attribute, or split a table behind a compatible view.
A NECESSARY CONDITION
Stable mappings
Views and mappings must remain compatible. Not every arbitrary logical change can be hidden from applications.
requires
Logical / conceptual schema
Entities or tables, attributes, relationships and constraints; independent of storage details.
can change with
preserves the logical schema
maps to storage structures
CHANGE THE STORAGE
Physical data independence
Change physical organization without changing the logical schema or applications. Example: add an index or reorganize files.
Internal / physical schema
Files, pages, indexes and storage organization.
can change with


DBA · GOVERNS & OPERATES
Database administrator
Manages security and authorization, schema and storage. Plans backup and recovery, monitors and tunes performance, and maintains availability.
BUILDS THE EXPERIENCE
Application developer
Designs applications, interfaces and business logic. Uses parameterized SQL or APIs; validates input and tests behavior. Works with the DBA on schema and access.
USES THE DATA
End user
Uses apps and reports. Enters, updates and queries data within permissions. Provides requirements and feedback to improve the application.
provides feedback
collaborates on schema & access
sees reports / results
operates; enters / queries
builds & tests
THE USER'S ACCESS PATH
User-facing application
Validates input, applies business logic and sends parameterized SQL / API calls. Presents results and reports.
administers & monitors
submits authorized requests
MANAGED BUSINESS DATA
Shared database
Persistent records and indexes. Users do not bypass the DBMS to access these files.
reads / writes
DATABASE MANAGEMENT SYSTEM
DBMS
Controls shared access, checks permissions, processes queries and manages transactions and storage.
returns results
returns data